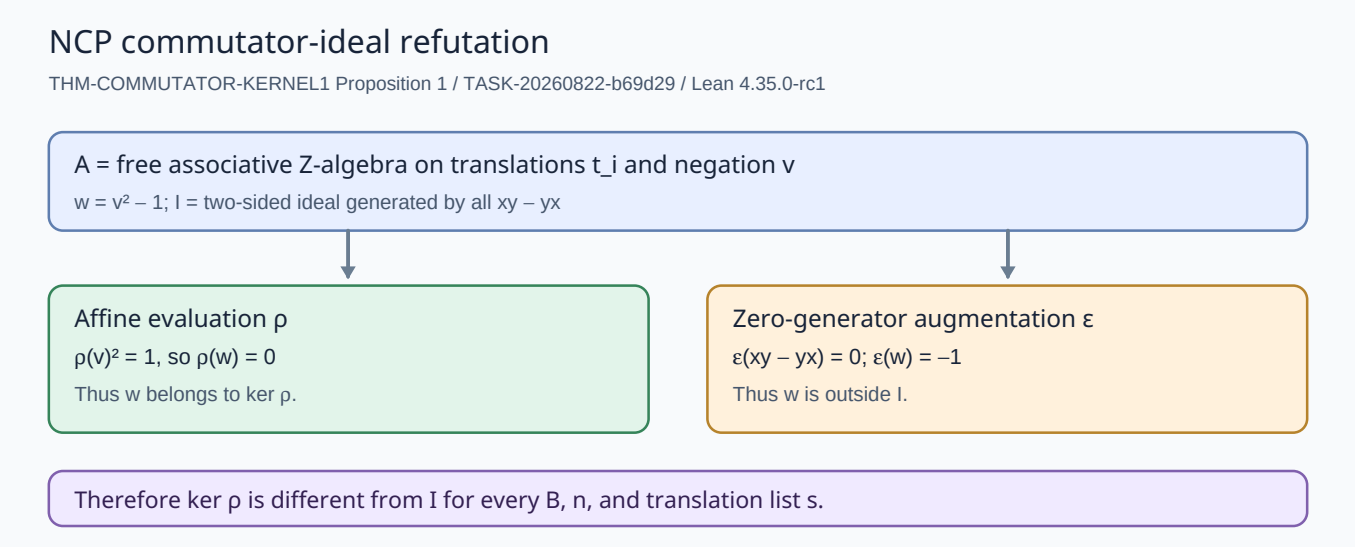
NCP commutator-ideal refutation

2026-10-21; THM-COMMUTATOR-KERNEL1 Proposition 1; TASK-20260822-b69d29

Scope. The free associative $\mathbb{Z}$-algebra A has generators $t_1, \ldots, t_B, v$. Let I be the two-sided ideal generated by all commutators $xy - yx$. Evaluation ρ sends t_i to translation by s_i and v to negation in the group algebra of the affine permutation group of $\mathbb{Z}/n\mathbb{Z}$. The Lean theorem proves $\ker \rho \neq I$ for every B, n, and translation list s.

Witness and separation

Set $w = v^2 - 1$. Negation is an involution even when $n = 1$ or 2, so $\rho(w) = 0$. The algebra homomorphism $\varepsilon : A \to \mathbb{Z}$ sending every free generator to zero kills all commutators, hence $I \subseteq \ker \varepsilon$. But $\varepsilon(w) = -1 \neq 0$. Therefore $w \notin I$ and $\ker \rho \neq I$. This argument separates the witness without constructing the whole commutative-polynomial quotient.



Validation and limits

Check	Observed result
Pinned Lean target	Machine verified on Lean 4.35.0-rc1; 2,077 build jobs passed; project audit passed 506 theorem declarations using standard axioms only.
Named theorem axioms	`propext`, `Classical.choice`, and `Quot.sound`; no `sorry` or extra axioms.
Complete target corpus	66 machine verified; zero failed, zero infrastructure failures, zero undischarged. All 66 receipts bind to the source commit; all require independent semantic review.

The proof is uniform; it therefore covers instances where the given list makes evaluation surjective. It does not itself establish surjectivity for arbitrary lists, the full kernel presentation, or algorithmic performance. Independent semantic review remains pending. No timing datum was produced, so canonical numerical dashboards are unchanged.